

## **Banking System ERD & Relational Schema**

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# SRS

## Banking System Description:

A bank needs a database system to manage its daily operations. The system should store information about customers, their accounts, loans, transactions, branches, and employees. Each customer has a unique customer ID, first name, last name, national ID, phone number, email, and address. A customer can own one or more accounts, while each account belongs to only one customer.

Each account has an account number, type, balance, and open date. Accounts are managed by bank branches, and each account is associated with exactly one branch. A branch has a branch name and phone number, and a branch may have multiple locations. Each branch also employs several employees. Every employee has an employee ID, name, position, and salary, and works in one branch.

Customers may also apply for loans. Each loan has a loan number, type, amount, interest rate, and start date, and it is associated with one customer. A loan can be paid through multiple payments. Each payment has a payment number, amount, and date, and it is related to a specific loan.

Accounts can perform financial transactions such as deposits and withdrawals. Each transaction has a transaction ID, amount, type, and date, and it is associated with one account.

There are two types of accounts in the system: savings accounts and checking accounts. A savings account has an interest rate, while a checking account has an overdraft amount.

